



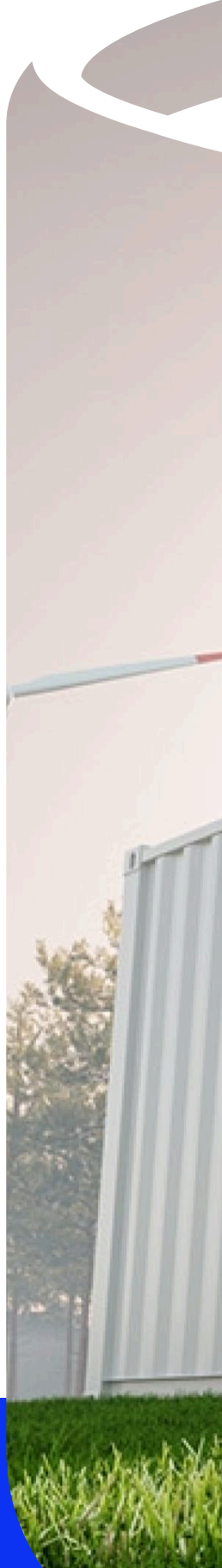
THE FUTURE OF ENERGY

Energy Storage Systems Products & Solutions Catalog



ABOUT US

Voltra Energy is shaping the future of energy in India with its advanced Battery Energy Storage Systems (BESS) with cutting-edge technology and world-class infrastructure. The company is focused on creating indigenized BESS solutions with state-of-the-art gigafactory and top-tier R&D for renewable energy integration, grid stabilization, and enhanced energy reliability across residential, commercial, and industrial sectors.





VOLT-LINK

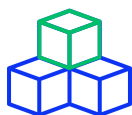
The liquid-cooled battery cabinet adopts advanced cabinet-level liquid cooling and temperature balancing strategy. The modular design makes the parallel solution more flexible and can be combined with the centralized PCS to form an ESS with higher energy density, which significantly improves the economy, safety and construction convenience of ESS projects

418 | 836 | 1254 | 1672 kWh



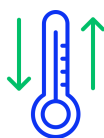
Safe and Reliable

In-PACK fire warning and protection with NOVEC1230/aerosol



Compact and Modular

1.7m² footprint only, easy transportation & fast installation.



Efficient Cooling

Optimal in-PACK duct design, achieve high-efficient cooling and low energy consumption



High Integration

Multiple units connected in parallel achieve MV/HV connection with PCS-boost containers.



Specifications

DC Side

Cell Type	LFP 314Ah Prismatic
PACK	52.24 kWh/1P52S
Battery System	418 kWh
Voltage Range	1165~1498Vdc
Rated Voltage	1331.2Vdc

AC Side

Rated Power	200 kW
Max. Power	220 kVA
THDi	≤ 3%
DC Ratio	0.5%I _{pn}
Nominal Voltage	690 Vac
Power Factor	-1 lagging -1 leading
Nominal Frequency	50 Hz/ 60Hz

General

Efficiency	94%
Charge/Discharge Rate	0.5C
DoD	95% (25±2°C)
25°C Cycle Life	≥ 8,000 times
Switching Time	100ms
Connectivity	Ethernet/RS485
Ingress Rating	IP55
Cooling	Liquid cooling
Operating Temperature	-25°C~55°C
Humidity	0~95%RH. non-condensing
Noise	80 dB
Altitude	≤ 2,000m (derating above 2,000m)
Fire Safety	NOVEC1230/Aerosol
Dimensions (W*D*H)	1,300*1,300*2,400 (mm)
Weight	~4,000 kg

VOLT-LINK AIR

The all-in-one air-cooled ESS cabinet integrates a long-life battery, efficient balancing BMS, high-performance PCS, active safety system, smart distribution, and HVAC into one cabinet, enabling long-term operation with safety, stability, and reliability. Through AC side parallel connection, it achieves flexible capacity expansion up to MWh.

112 | 241 | 482 | 964 kWh



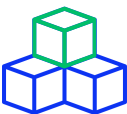
Economical and Efficient

Conversion efficiency over 90%,
DoD over 96%



Safe and Reliable

IP55 protection, optimized ventilation,
cells temp. difference $\leq 6^{\circ}\text{C}$



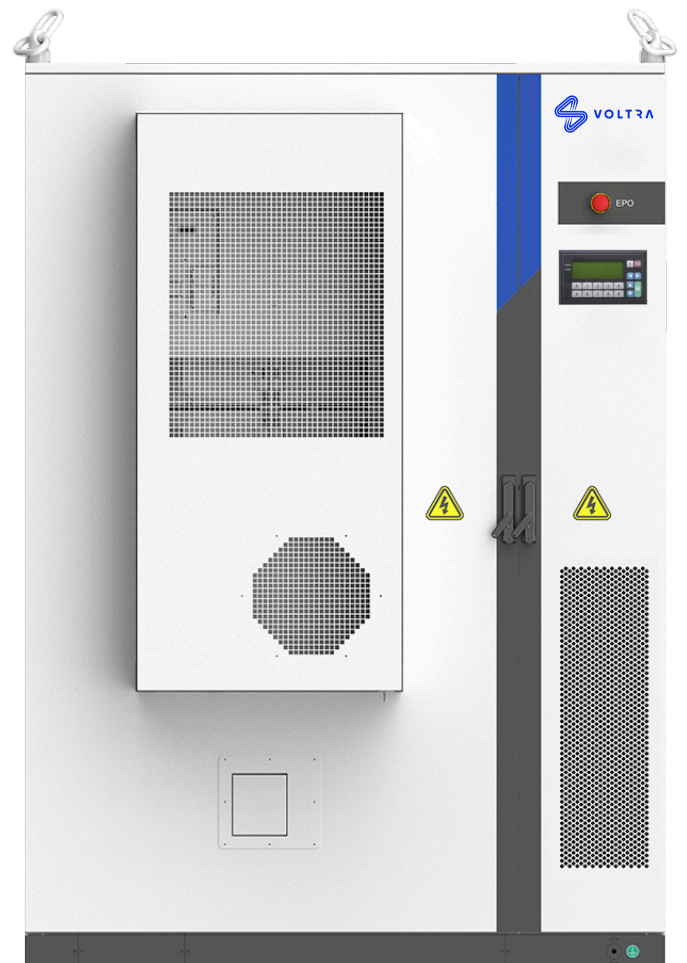
Compact and Modular

1.6 m² footprint, modular,
simplified parallel expansion



Smart O&M

Diversified O&M access, both
on app and Cloud



Specifications

DC Side

Cell Type	LFP 314Ah
PACK	16.07 kWh/1P16S
Battery System	241 kWh/1P240S
Voltage Range	672 – 876 Vdc
Rated Voltage	768 Vdc

AC Side

Rated Power	125 kW
Max. Power	135 kW
THDi	≤ 3%
DC Ratio	0.5% pn
Nominal Voltage	400 Vac/3P+N+PE
Power Factor	-1 lagging -1 leading
Nominal Frequency	50 Hz/60 Hz

General

Efficiency	≥ 90%
Charge/Discharge Rate	0.5 P
DoD	95% (25±2°C)
25°C Cycle Life	≥ 8,000 times
Switching Time	100ms
Connectivity	Ethernet/RS485
Ingress Rating	IP55
Cooling	Forced air cooling
Operating Temperature	-25°C-55°C
Humidity	0-95%RH. non-condensing
Noise	80 dB
Altitude	≤ 2,000m (derating above 2,000m)
Fire Safety	Aerosol
Dimensions (W*D*H)	1,430*1,270*2,300 (mm)
Weight	~2,600 kg

VOLT-HVC

IP54

Voltra high voltage series uses a 5 KWH standard modular design, with multi-module in series and support multi-cluster in parallel. It's suitable for various application scenarios such as villas, farms, and small C&I power supplies, and provides a complete set of green, low-carbon, and reliable energy solutions.

30.72 | 40.96 | 51.2 | 61.44 kWh



Ultra-Long Life

Design life of 15 years, can reach up to 6,000 cycles



Intelligent Management

Three-level architecture, identify the number and address of modules



Safe and Reliable

Multi-layer protection with BMS and thermal monitoring.



Flexible Configuration

Multi-module series, to meet the requirements



Specifications

Model	Model - 1	Model - 2	Model - 3	Model - 4
				
Battery Module	HVC05 [5.12 kWh]			
Cell Type	100Ah LFP Prismatic Cell			
Number of Modules	6	7	8	10
System Nominal Energy	30.72kWh	35.84kWh	40.96kWh	51.2kWh
Available System Nominal Energy	29.184kWh	34.048kWh	38.912kWh	48.64kWh
Nominal Voltage	256.0V	358.4V	409.6V	512V
Max. Output Power	30.72kW	35.84kW	40.96kW	51.2kW
Operating Voltage Range	268.8~345.6V	313.6~403.2V	358.4~460.8V	448~576V
Max. Charging Current	50A	50A	50A	50A
Max. Discharge Current	100A	100A	100A	100A
General				
Dimensions (W*D*H)	2100*1750*750mm	2100*1750*750mm	2100*1750*750mm	2100*1750*750mm
Weight	472kg	524kg	576kg	680kg
Communication	CAN			
Installation	Floor			
Enclosure Rating	IP54			
25°C Cycle Life	≥6000Cycles @ 90% DOD & 70% EOL			
Operating Temperature	Charge: -5~50°C; Discharge: -20~50°C; Recommended: 15~35°C			
Operating Humidity	5%~95% [Non-Condensing]			
Cooling Mode	Natural Cooling			
Operating Altitude	≤4000m			

VOLT-LV314

This system is a compact, efficient, and space-saving solution for storing electrical energy, typically used in residential or small commercial applications. These systems are rack mounted and stack mounted and are designed to optimize energy usage, improve power reliability, and allow integration with solar panels. Can support a maximum of up to 16 batteries in parallel expanding to 256 kWh.

16.07 | 32.14 | 48.21 | 64.28 kWh



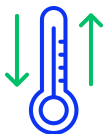
Flexible and Compatible

16kWh modular design,
support 1-16 in parallel



Easy Installation

floor mounted, save
installation time & cost



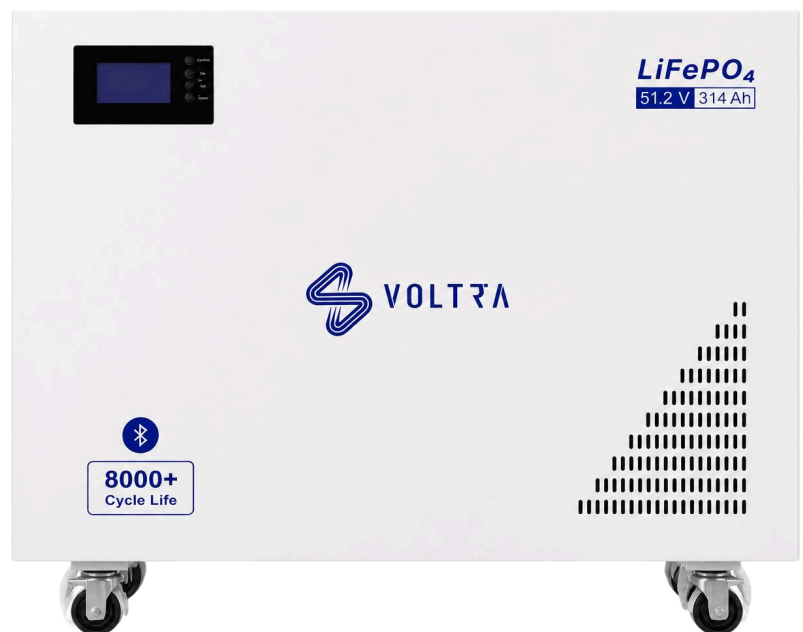
Environmental Adaptability

Wider temperature range:
(-20°C ~ 55°C)



Long Lifespan

More than 15 years designed
lifespan, >8000 cycles (0.5C, 25°C)



Specifications

Battery Module	LV314
Cell Type	LFP
Nominal Voltage	51.2V
Operating Voltage Range	44.8-57.6V
Number of Modules	1
Available Energy	16.07kWh
Max. Output Power	9.72kW
Peak Output Power	11.26kW, 10s
Max. Charging Current	200A
Max. Discharge Current	200A
Recommended Charge Current	100A

General

Dimensions (W*D*H)	700*250*520(mm)
Weight	145kg (With Packaging)
Communication	CAN/RS485
Installation	Floor Mounted
Operating Temperature	Charge: -5~50°C; Discharge: -20~50°C; Recommended: 15~35°C
Operating Altitude	≤4000m
Scalability	1~16
Display	LED
Terminal	M8
25°C Cycle Life	>8000 (0.5C Discharge @ 25°C, 70% EOL)

VOLT-LV200

This floor-mounted system is a compact, efficient, and space-saving solution for storing electrical energy, typically used in residential or small commercial applications. These systems are installed on floor, either inside or outside a building, and are designed to optimize energy usage, improve power reliability, and allow integration with renewable energy sources such as solar panels. Can support a maximum of up to 16 batteries in parallel and the capacity can be expanded to 164 kWh.

10.2 | 20.4 | 40.8 | 81.6 kWh



Flexible and Compatible

10.24kWh modular design,
support 1-16 in parallel



Easy Installation

Wall mounted/floor mounted,
save installation time & cost



Environmental Adaptability

Wider temperature range:
(-20°C ~ 55°C)



Long Lifespan

More than 15 years designed
lifespan, >6000 cycles (0.5C, 25°C)



Specifications

Battery Module	LV16S2P
Cell Type	LFP
Nominal Voltage	51.2V
Operating Voltage Range	44.8~57.6V
Number of Modules	1
Available Energy	10.24kWh
Max. Output Power	10kW
Peak Output Power	10.8kW, 20s
Max. Charging Current	100A
Max. Discharge Current	200A
Charge-Discharge Efficiency	≥97%

General

Dimensions (W*D*H)	510*175*1025(mm)
Weight	115 Kg (With Packaging)
Communication	CAN/RS485
Installation	Floor Standing
Operating Temperature	Charge: -5~50°C; Discharge: -20~50°C; Recommended: 15~35°C
Operating Altitude	≤4000m
Scalability	1~16
25°C Cycle Life	>6000 (0.5C Discharge @ 25°C, 70% EOL)

VOLT-LVS

It adopts an industrial aesthetic and N+1 stackable design. The system is composed of 100Ah modules, supports a maximum of 16 parallel groups, and the capacity can be expanded to 82kWh. Matched with mainstream brands of inverters, the system can be applied to grid-connected, off-grid, photovoltaic, and all kinds of green and low-carbon applications of household ESS.

5.1 | 10.2 | 15.3 | 20.4 kWh



Wide Compatibility

Matches 20+ brands of mainstream inverters



High Security

Self-produced 100Ah LFP prismatic cell, highly safe



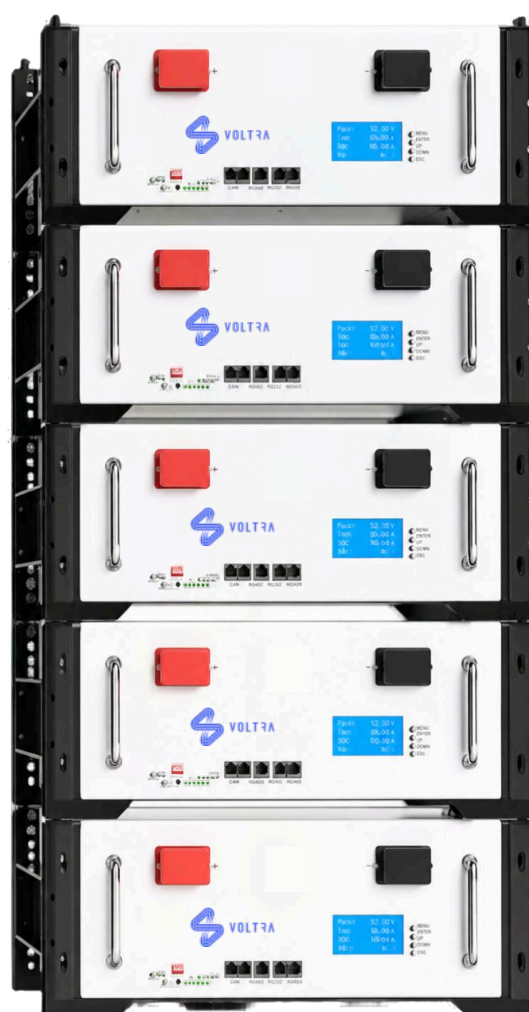
Ultra-Long Life

Design of 15 years, can reach up to 6,000 cycles



High Scalability

Optional expansion functions like anti-theft, fire protection, etc.



Specifications

Model	MODEL 1	MODEL 2	MODEL 3	MODEL 4
				
Battery Module	LVS 05			
Cell Type	100Ah LFP Prismatic Cell			
Number of Modules	1	2	3	4
Nominal Voltage	51.2V			
Operating Voltage Range	44.8-57.6V			
System Nominal Energy	5.12kWh	10.24kWh	15.36kWh	20.48kWh
Available System Nominal Energy	4.86kWh	9.72kWh	14.6kWh	19.45kWh
Max. Output Power	5kW	10kW	15kW	20kW
Peak Output Power	5.4kW, 10s	10.5kW, 10s	15.5kW, 10s	21.0kW, 10s
Max. Charging Current	50A	100A	150A	200A
Max. Discharge Current	100A	200A	300A	400A
Charge and Discharge Efficiency	≥97%			
General				
Dimensions (W*H*D)	425*250*573mm	425*500*573mm	425*750*573mm	425*1000*573mm
Weight	60kg	120kg	180kg	240kg
Communication	CAN/RS485			
Installation	Stack/Stand			
Enclosure Rating	IP21			
25°C Cycle Life	≥3500 Cycles @ 90% DOD & 70% EOL			
Operating Temperature	Charge: -5~50°C; Discharge: -20~50°C; Recommended: 15~35°C			
Operating Humidity	5%-95% [Non-Condensing]			
Cooling Mode	Natural Cooling			
Operating Altitude	≤4000m			

VOLT-LVW 16S

IP65

This wall-mounted system is a compact, efficient, and space-saving solution for storing electrical energy, typically used in residential or small commercial applications. These systems are installed on a wall, either inside or outside a building, and are designed to optimize energy usage, improve power reliability, and allow integration with renewable energy sources such as solar panels. Can support a maximum of up to 16 batteries in parallel and the capacity can be expanded to 82 kWh.

5.12 | 10.24 | 20.48 | 40.96 kWh



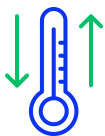
Flexible and Compatible

5.12kWh modular design,
support 1-16 in parallel



Easy Installation

Wall mounted/floor mounted,
save installation time & cost



Environmental Adaptability

Wider temperature range:
(-20°C ~ 55°C)



Long Lifespan

More than 10 years designed
lifespan, >3500 cycles (0.5C, 25°C)



Specifications

Battery Module	LV16S
Cell Type	LFP
Nominal Voltage	51.2V
Operating Voltage Range	44.8~57.6V
Number of Modules	1
Available Energy	5.12kWh
Max. Output Power	5kW
Peak Output Power	5.4kW, 10s
Max. Charging Current	50A
Max. Discharge Current	100A
Charge-Discharge Efficiency	≥97%

General

Dimensions (W*H*D)	559*500*173mm
Weight	60kg
Communication	CAN/RS485
Installation	Floor/Wall Hanging
Operating Temperature	Charge: -5~50°C; Discharge: -20~50°C; Recommended: 15~35°C
Operating Altitude	≤4000m
Scalability	1~16

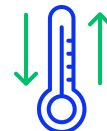
VOLT-LVW 8S **PRO**

This wall-mounted system is a compact, efficient, and space-saving solution for storing electrical energy, typically used in residential or small commercial applications. These systems are installed on a wall and are designed to optimize energy usage, improve power reliability, and allow integration with solar panels.



Easy Installation

Wall mounted/floor mounted,
save installation time & cost



Environmental Adaptability

Wider temperature range:
(-20°C ~ 55°C)



Long Lifespan

More than 10 years designed
lifespan, >3500 cycles (0.5C, 25°C)

Specifications

Battery Module	LV8S2P
Cell Type	LFP
Nominal Voltage	25.6V
Nominal Capacity	200Ah
Operating Voltage Range	19.2-28.8V
Number of Modules	1
Available Energy	5.12kWh
Max. Output Power	3.6kW
Peak Output Power	4kW, 10s
Max. Charging Current	100A
Max. Discharge Current	150A
Charge-Discharge Efficiency	≥97%

General

Dimensions	550 × 500 × 170 mm
Weight	55kg
Installation	Floor/Wall Hanging
Operating Temperature	Charge: -5~50°C; Discharge: -20~50°C; Recommended: 15~35°C
Operating Altitude	≤4000m

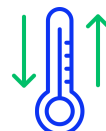
VOLT-LVW 8S

This wall-mounted system is a compact, efficient, and space-saving solution for storing electrical energy, typically used in residential or small commercial applications. These systems are installed on a wall and are designed to optimize energy usage, improve power reliability, and allow integration with solar panels.



Easy Installation

Wall mounted/floor mounted,
save installation time & cost



Environmental Adaptability

Wider temperature range:
(-20°C ~ 55°C)



Long Lifespan

More than 10 years designed
lifespan, >3500 cycles (0.5C, 25°C)

Specifications

Battery Module	LV8S
Cell Type	LFP
Nominal Voltage	25.6V
Operating Voltage Range	19.2-28.8V
Number of Modules	1
Available Energy	2.56kWh
Max. Output Power	2.4kW
Peak Output Power	2.7kW, 10s
Max. Charging Current	50A
Max. Discharge Current	100A
Charge-Discharge Efficiency	≥97%

General

Dimensions	490*281.7*170 (mm)
Weight	28kg
Installation	Floor/Wall Hanging
Operating Temperature	Charge: -5~50°C; Discharge: -20~50°C; Recommended: 15~35°C
Operating Altitude	≤4000m

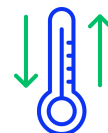
VOLT-LVW 4S PRO

This wall-mounted system is a compact, efficient, and space-saving solution for storing electrical energy, typically used in residential or small commercial applications. These systems are installed on a wall and are designed to optimize energy usage, improve power reliability, and allow integration with solar panels.



Easy Installation

Wall mounted/floor mounted,
save installation time & cost



Environmental Adaptability

Wider temperature range:
(-20°C ~ 55°C)



Long Lifespan

More than 10 years designed
lifespan, >3500 cycles (0.5C, 25°C)

Specifications

Battery Module	LV4S2P
Cell Type	LFP
Nominal Voltage	12.8V
Operating Voltage Range	10.4V – 14.4V
Number of Modules	1
Available Energy	2.56kWh
Max. Output Power	2.4kW
Peak Output Power	2.7kW, 10s
Max. Charging Current	50A
Max. Discharge Current	100A
Charge-Discharge Efficiency	≥97%

General

Dimensions	490*281.7*170 (mm)
Weight	28kg
Installation	Floor/Wall Hanging
Operating Temperature	Charge: -5~50°C; Discharge: -20~50°C; Recommended: 15~35°C
Operating Altitude	≤4000m

VOLT-LVW 4S

This wall-mounted system is a compact, efficient, and space-saving solution for storing electrical energy, typically used in residential or small commercial applications. These systems are installed on a wall and are designed to optimize energy usage, improve power reliability, and allow integration with solar panels.



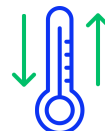
Easy Installation

Wall mounted/floor mounted,
save installation time & cost



Long Lifespan

More than 10 years designed
lifespan, >3500 cycles (0.5C, 25°C)



Environmental Adaptability

Wider temperature range:
(-20°C ~ 55°C)

Specifications

Battery Module	LV4S
Cell Type	LFP
Nominal Voltage	12.8V
Operating Voltage Range	9.6-14.4V
Number of Modules	1
Available Energy	1.25kWh
Max. Output Power	1.2kW
Peak Output Power	1.35kW, 10s
Max. Charging Current	50A
Max. Discharge Current	100A
Charge-Discharge Efficiency	≥97%

General

Dimensions (W*H*D)	282*282*175 (mm)
Weight	15kg
Installation	Floor/Wall Hanging
Operating Temperature	Charge: -5~50°C; Discharge: -20~50°C; Recommended: 15~35°C
Operating Altitude	≤4000m

VOLTRA TEST LABS



Cell Grading

Combined with the cell failure mechanism model, it monitors all cells with charge and discharge in real-time.



Container Testing

Multiple cycles are tested to comply with safety standards and monitoring communication & control systems for all the components.



BMS Testing

Each BMS is tested with detailed parameters like voltage, temperature accuracy, balancing, over-under voltage, voltage interlock, insulation, shunt accuracy etc.



Insulation Test

To verify the integrity of the insulation in the battery pack and its components, to ensure no electrical leakage paths that could pose safety risks.



Pack Grading

Each battery pack is rigorously tested with EU standards to check capacity, voltage, charge-discharge, safety, cut-off parameters, etc.



Environmental Testing

Exposes the battery under different environmental conditions, such as temperature extremes, humidity, pressure, vibration, and other external factors.



Pressure Leak Test

The test verifies that the liquid cooling system is sealed and free of leaks, preventing coolant loss, which could lead to overheating or thermal management failures.



OCV-IR Testing

It is the fastest and most effective way to judge battery health, safety, and suitability for use before the cell is put into service or assembled into a pack.



Push-Pull Tester

The push-pull test is a mechanical strength test used to verify the quality and reliability of laser welds in lithium battery cells, modules, and packs—especially at tab-to-busbar.

The VOLTRA Gigafactory

Voltra BESS is proud to operate a cutting-edge gigafactory dedicated to the production of state-of-the-art Battery Energy Storage Systems (BESS). Our facility, located in a prime industrial zone, incorporates the latest automation, efficiency, and sustainability standards. It is designed to meet the increasing demand for high-performance energy storage solutions in renewable energy applications.

Voltra's gigafactory is central to the global shift towards sustainable energy, utilizing scalable production methods and green energy. With advanced robotics and AI-driven processes, we ensure efficient production while minimizing environmental impact.



Energy Saving and Fast

1P fast charge/discharge rate,
energy storing & releasing



Economical and Efficient

Conversion efficiency over 90%,
DoD over 96%



Smart O&M

Diversified monitoring by HMI
(local), app/web (remote)



Safe and Reliable

IP55, thermal management, cell
difference $\leq 6^{\circ}\text{C}$



ROBUST
TESTING



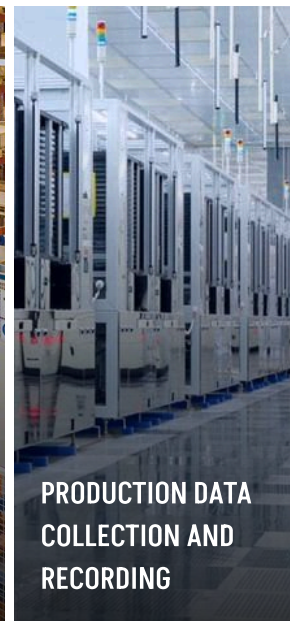
TROUBLE-
SHOOTING AND
EARLY WARNING



INTELLIGENT
PRODUCTION
LINE



TECHNOLOGY
INTEGRATION AT
EVERY STEP



PRODUCTION DATA
COLLECTION AND
RECORDING



MAKE IN INDIA

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